

Quiz: defining data

This quiz has 5 questions. The passing grade is 4. [Back](#) [Next](#)

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State	Finished
Completed on	Wednesday, January 15, 2025, 2:22 PM
Time taken	19 mins 39 secs
Grade	1.00 out of 5.00 (20%)

Question **1**

Incorrect

0.00 points out of 1.00

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Which of these definitions reserve a single byte containing X'C1' (EBCDIC 'A')? (Select 3)

Select one or more:

- ☐ A. DC BL1'1100 0001 0000 0000'
- ☐ B. DC CL1'AB'
- ☐ C. DC CL1'BA'
- ☒ D. DC XL1'00C1' ✓
- ☒ E. DC BL1'0000 0000 1100 0001' ✓
- ☒ F. DC XL1'C100' ✗

When the value is greater than the explicit length, character fields are truncated on the right, and hexadecimal and binary fields are truncated on the left.

The correct answers are:

DC CL1'AB'

This is truncated on the right, leaving C'A'.

DC XL1'00C1'

This is truncated on the left, leaving X'C1'.

DC BL1'0000 0000 1100 0001'

This is the same as X'00C1'. It is truncated on the left, leaving B'1100 0001' (X'C1').

The incorrect answers are:

DC XL1'C100'

Truncated on the left, this generates X'00'.

DC CL1'BA'

Truncated on the right, this leaves C'B'.

DC BL1'1100 0001 0000 0000'

Truncated on the left, this leaves B'0000 0000'.

The correct answers are:

DC CL1'AB'

,

DC XL1'00C1'

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,

DC BL1'0000 0000 1100 0001'

Question 2

Incorrect

0.00 points out of 1.00

Select the statement that defines a **doubleword binary integer** containing **decimal 77000**

Select one:

- ☐ A. DC H'77000'
- ☐ B. DC F'77000'
- ☒ C. DC D'77000' ✖
- ☐ D. DC FD'77000'
- ☐ E. DC C'77000'

DC FD'77000' is the correct answer (the "D" type modifier specifies a doubleword)

DC D'77000' defines a floating-point number, not a binary integer

DC H'77000' is an invalid statement, as the maximum positive halfword value is 32767

DC F'77000' defines a fullword

DC C'77000' defines a 5-byte character field

The correct answer is: DC FD'77000'

Question **3**

Correct

1.00 points out of 1.00

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DEFINE CONSTANT (DC) and DEFINE STORAGE (DS) can specify these data types (Select 4)

Select one or more:

- ☐ A. Computer instructions
- ☒ B. Binary integers, such as fullwords or halfwords ✓
- ☒ C. Hexadecimal digits ✓
- ☒ D. Bits ✓
- ☒ E. Characters ✓

Correct!

DC and DS can specify these data types:

Binary integers, such as fullwords or halfwords: types **H, F, FD**

Characters: type **C**

Bits: type **B**

Hexadecimal digits: **X**

Neither DC nor DS can be used to define computer instructions.

The correct answers are: Binary integers, such as fullwords or halfwords, Characters, Bits, Hexadecimal digits

Question **4**

Incorrect

0.00 points out of 1.00

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Which of these fields is placed by the Assembler on a **fullword** alignment? (Select 3)

Select one or more:

- ☐ A. Fld2 DC F'100'
- ☐ B. Fld1 DC FD'-50'
- ☒ C. Fld4 DC FL4'1000' ✕
- ☐ D. Fld5 DC 4F'0'
- ☐ E. Fld3 DC H'-200'

The correct answers are:

Fld2 DC F'100'
Fld5 DC 4F'0'

These are fullword-aligned, as defined.

Fld1 DC FD'-50'

This is a doubleword-aligned field, so it is also fullword-aligned.

The incorrect answers are:

Fld3 DC H'-200'

This is halfword aligned. A halfword, at a multiple-of-2 address, may or may not also be at a fullword (multiple-of-4) address.

Fld4 DC FL4'1000'

Although this is a fullword, it is not placed on a fullword alignment, because it has an explicit length.

The correct answers are:

Fld1 DC FD'-50'
,
Fld2 DC F'100'

,
FId5 DC 4F'0'

Question **5**

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Incorrect

0.00 points out of 1.00

Which of these is not a valid Assembler statement?

Select one:

- ☐ A. DS F
- ☒ B. DS F'15000' ✖
- ☐ C. DS 0F
- ☐ D. DC F
- ☐ E. DC 0F

This statement is invalid...

DC F

...because the value is missing. Value is mandatory when the duplication factor is not zero (in this case, it is 1, by default).

DC 0F

is valid because the duplication factor is zero, so a value is not required.

The remaining statements are DEFINE STORAGE (**DS**). For DS, a value is always optional. They are all valid Assembler statements.

The correct answer is:

DC F